

# Eurostat data analysis

## Geospatial data management project

Federico Garegnani

Università degli Studi di Milano

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# Presentation of the datasets used (1)

3 datasets:

## ① *Census grid 2021*

- Grid of  $1 \text{ km}^2$  cells
- Columns: sex, age group, activity status, place of birth, population, population one year prior the census

## ② *Geographic accessibility*

- Grid of cells of  $1 \text{ km}^2$
- Accessibility as travel time from each cell to the nearest healthcare service
- To catch each grid cell to the road network, the nearest network node from the centroid inside the cell was used
- Time measured as travel by car using the road network, data from TomTom Multinet datasets. Dijkstra algorithm used to calculate the shortest distance



# Presentation of the datasets used (2)

## 3 *Communes*

- Administrative boundaries of municipalities
- Columns: commune id, country id, country code, commune name, ASCII name, true flag, NSI code, NSI name, Latin name, NUTS code

All the datasets use ETRS89 as reference system.



# Filtering the datasets

Definition of table `it_communes` and indexes for all the tables.

```
CREATE TABLE it_communes AS (  
    SELECT *  
    FROM "all_communes" c  
    WHERE "CNTR_CODE" = 'IT'  
);  
  
CREATE INDEX it_communes_idx  
ON it_communes USING GIST (geom);  
...
```



# Calculation of the population by municipality

The formula used:

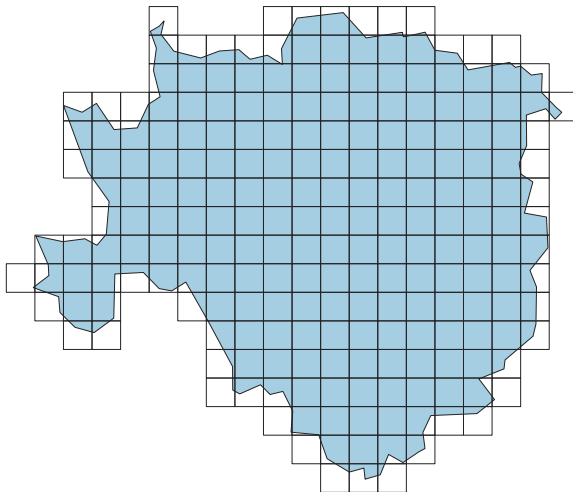
$$P_C = P_G \sum \frac{A(C \cap G)}{A(G)} \quad (1)$$

and the query performed:

```
WITH pop_calc AS (  
  SELECT  
    c."COMM_ID",  
    ROUND(SUM(ST_Area(ST_Intersection(g.geom, c.geom)) / 1000000  
      ↪ * g."T")) AS pop  
  FROM census_2021 g JOIN it_communes c ON ST_Intersects(g.geom,  
    ↪ c.geom)  
  GROUP BY c."COMM_ID"  
)  
UPDATE it_communes c  
SET pop = p.pop  
FROM pop_calc p  
WHERE c."COMM_ID" = p."COMM_ID";
```



# Overlapping of polygons and grid cells



**Figure:** Polygon representing the commune of Milan overlapped with the grid cells intersecting it



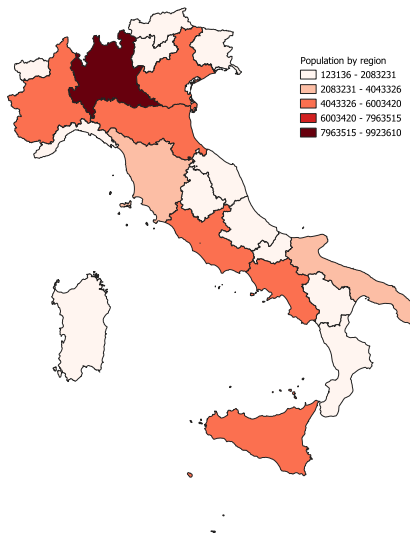
# Calculation of mean accessibility value

Computation of the mean accessibility times by commune to the closest healthcare structure and to the 3 closest ones. Years 2020 and 2023.

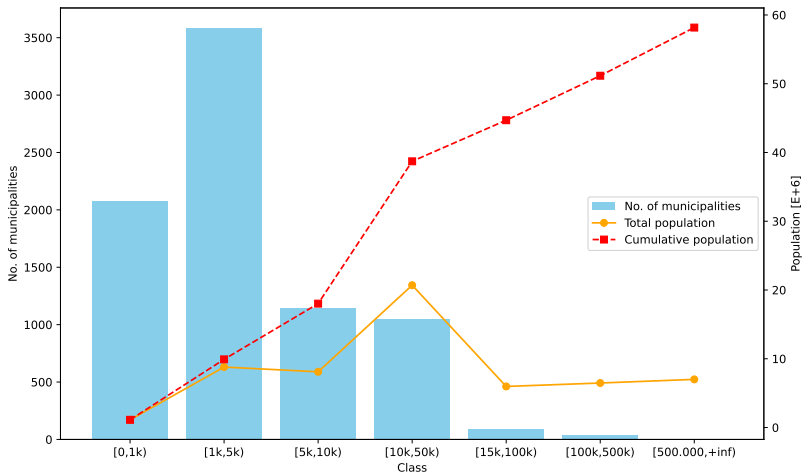
```
WITH avg_t AS (  
  SELECT  
    c."COMM_ID",  
    AVG(g.health_2020_n1) AS mean_health_2020_n1,  
    AVG(g.health_2020_n3) AS mean_health_2020_n3,  
    ...  
  FROM it_communes c JOIN grid_accessibility_health g ON  
    ↪ st_intersects(c.geom, g.geom)  
  GROUP BY c."COMM_ID"  
)  
UPDATE it_communes as c  
SET  
  mean_health_2020_n1 = ROUND(t.mean_health_2020_n1::NUMERIC, 2),  
  ...  
FROM avg_t t  
WHERE t."COMM_ID"=c."COMM_ID";
```



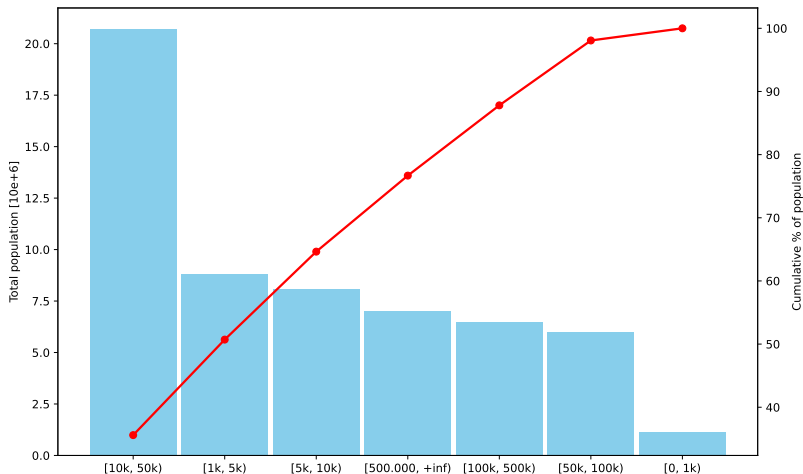
# Population by region



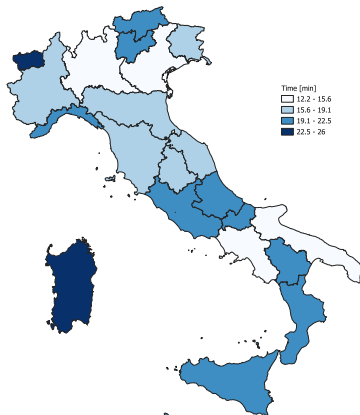
# Number of communes by size



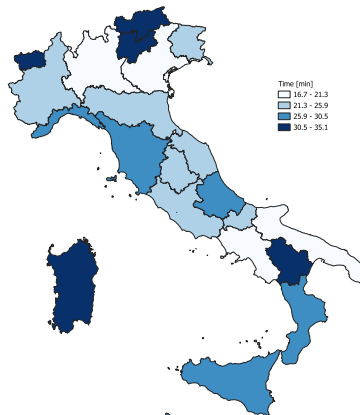
# Italian population by size of municipality



# Accessibility to healthcare services in Italy



(a) Acc. to the closest centre



(b) Acc. to the 3 closest centres



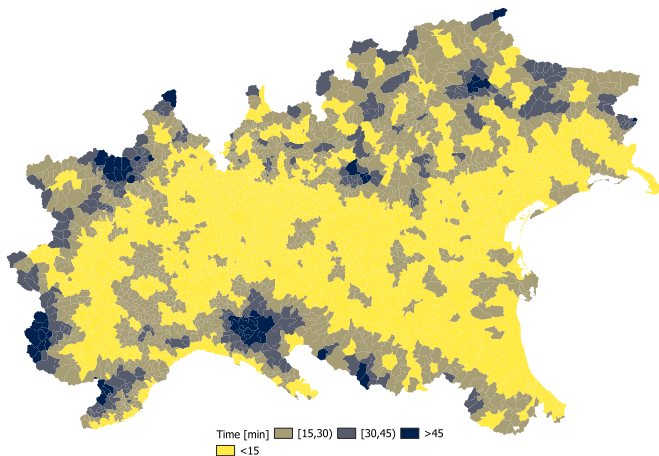
# Categories based on travel time

Access time divided in 4 categories:

| No. | Travel time<br>[min] | Accessibility |
|-----|----------------------|---------------|
| 1   | < 15                 | High          |
| 2   | [15, 30)             | Medium-high   |
| 3   | [30, 45)             | Medium-low    |
| 4   | > 45                 | Low           |



# Accessibility in the North of Italy (1)



**Figure:** Accessibility to the closest healthcare service



# Accessibility in the North of Italy (2)

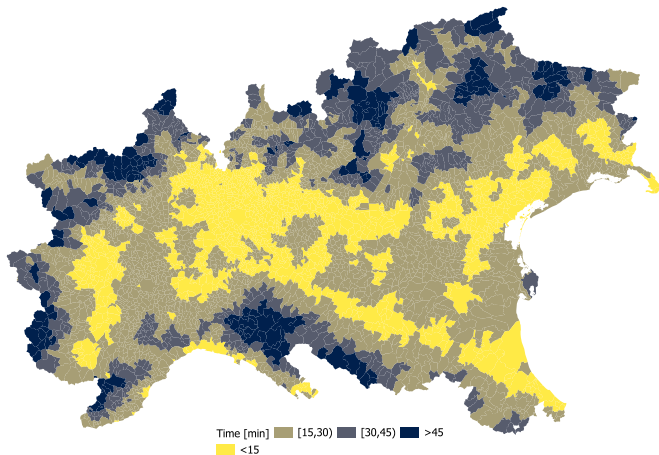


Figure: Accessibility to the 3 closest healthcare services



# Distribution of municipalities by class

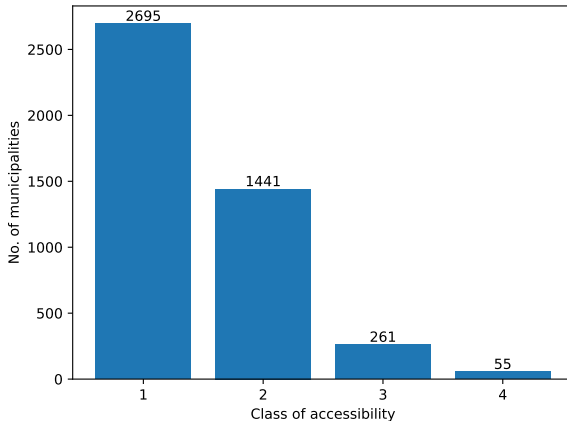


Figure: Time to reach the closest healthcare centre



# Population in low accessibility areas

Share of population by accessibility area:

| Class | Travel time | Population | % of total |
|-------|-------------|------------|------------|
| 1     | < 15        | 18 836 153 | 69.32      |
| 2     | [15, 30)    | 7 399 725  | 27.23      |
| 3     | [30, 45)    | 792 058    | 2.91       |
| 4     | > 45        | 143 979    | 0.53       |
|       |             | 27 171 915 |            |



# Change in healthcare accessibility

7 intervals of change have been defined:

| Class                | Interval     | Communes |       |
|----------------------|--------------|----------|-------|
|                      | %            | No.      | %     |
| Sharp improvement    | $\leq -20$   | 135      | 3.03  |
| Relevant improvement | $(-20; -10]$ | 348      | 7.82  |
| Slight improvement   | $(-10; -5]$  | 586      | 13.16 |
| No change            | $(-5; 5)$    | 2298     | 51.62 |
| Slight worsening     | $[5; 10)$    | 494      | 11.1  |
| Relevant worsening   | $[10; 20)$   | 355      | 7.97  |
| Sharp worsening      | $\geq 20$    | 236      | 5.3   |
|                      |              | 4452     |       |



# Variation between 2020 and 2023

Computed with this formula:

$$\Delta_{n1}[\%] = \frac{\bar{t}_{2023n1} - \bar{t}_{2020n1}}{\bar{t}_{2020n1}} * 100$$

where

$\Delta$  Variation as percentage

$\bar{t}$  Mean access time in the specified year

Thus:

$\Delta > 0 \Rightarrow$  worsening

$\Delta < 0 \Rightarrow$  improvement



# Formula implementation in SQL

The query executed to compute accessibility variation between 2020 and 2023:

```
SELECT
  c."COMM_ID",
  c."COMM_NAME",
  c.mean_health_2020_n1,
  c.mean_health_2023_n1,
  ROUND(((c.mean_health_2023_n1 - c.mean_health_2020_n1) /
    ↪ c.mean_health_2020_n1 * 100.0)::NUMERIC, 2) AS
    ↪ variation_n1,
  c.geom
FROM it_communes c
WHERE SUBSTRING(c."NUTS_CODE" FROM 1 FOR 3) IN ('ITC', 'ITH');
```



# Map of changes

